Alt Text Generator | Tool Summary

Version: 1.0 (Stable)

Tech Stack: HTML5, TailwindCSS, Google Gemini 1.5 Flash, SheetJS (XLSX)

1. Executive Summary

The **Alt Text Generator** is a browser-based productivity tool designed to automate the creation of high-quality descriptive text for image assets. By leveraging Multimodal AI (Google Gemini), the tool analyzes image pixels to suggest multiple variations of alt text, allowing users to edit, select, and export their choices into a structured campaign document in seconds.

2. Key Capabilities

- **Batch Image Processing:** Unlike single-image tools, this application allows users to drag and drop multiple files at once, processing them as independent threads for maximum efficiency.
- **Variety-Based Suggestions:** For every image, the AI generates three unique descriptive variations, providing options that range from concise to highly detailed.
- **Live Inline Editing:** Users can refine the AI's output directly within the tool using a custom `contenteditable` interface, ensuring the final text perfectly matches the project's brand voice.
- **Real-Time Character Tracking:** Features a live character counter (125-character threshold) that alerts users if an edited description exceeds standard accessibility or SEO best practices.
- **Selection Validation:** The tool includes a "failsafe" audit logic that highlights any images missing a selection, preventing incomplete data from being exported.

3. Workflow Overview

1. **Upload Assets:** User drags and drops one or more image files into the browser-based drop zone.
2. **AI Analysis:** The tool uses Gemini 1.5 Flash to "read" the visual content and generate three text options per asset.
3. **Review & Refine:** User reviews the suggestions, edits them if necessary, and clicks their preferred version to highlight it in green.
4. **Session Audit:** Before exporting, the tool verifies that every uploaded image has a confirmed selection.
5. **Campaign Export:** User downloads a `.xlsx` file containing a table of filenames and their corresponding selected alt text.

4. Technical Specifications

Feature	Specification
Processing Engine	Client-Side JavaScript (No image data is stored on a server)
AI Model	Google Gemini 1.5 Flash (via Direct API Fetch)
Resiliency Logic	Automatic Exponential Backoff (Handles 429/503 Quota errors)
Export Format	Excel (.xlsx) via SheetJS
Security	API Keys are used locally; Data is transient (cleared on refresh)